HW1_search

Thursday, October 2, 2025 3:52 PM

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CS 331 **Introduction to Artificial Intelligence** Fall 2025

OREGON STATE UNIVERSITY

School of Electrical Engineering and Computer Science

Instructor: Sandhya Saisubramanian

Total points: 100 HW 1: **Agents, Uninformed Search, Informed Search** Due date: Oct 10, 2025

Instructions: This homework assignment consists of a written portion and a programming portion. Collaboration is not allowed on any part of this assignment. Solutions must be typed (hand written and scanned submissions will not be accepted) and saved as a .pdf file. You will submit a single .zip file that contains the code base and solutions as a .pdf file.

1. **(10 points)** Consider an autonomous vacuum cleaner agent that is equipped with sensors to detect dirt, walls, and battery level. Which agent architecture (simple reflux, model-based, goal-based, utility-based) is best suited for maximizing battery life while still cleaning effectively, and why?

2. **(20 points)** For the graph in Figure 1, implement **Breadth First Search** (BFS) and complete the table below with the order in which the nodes will be expanded. Include the shortest path found by BFS. Also include a screenshot of your code output in this document, in addition to filling in the table below. You are provided with a skeleton code in Python. You can also code it up your own way or in another language of your choice.

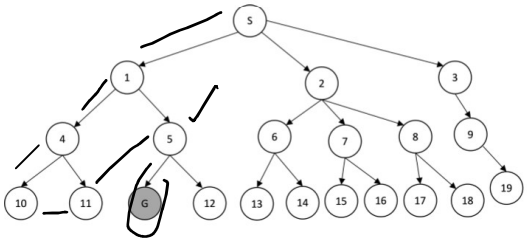

Figure 1:

Question 1:
I would say utility based because to maximize battery life and clean effectively, you'd want to find the "shortest path" through the rooms that are dirty and to only clean/activate suction when dirt is detected. You will have to optimize tradeoffs as there are many different paths to accomplish the goal of cleaning the house while saving battery life considering its environment is not fully observable.

```
BFS Expansion Order:
S -> 6 -> 7 -> 8 -> 1 -> 2 -> 3 -> 6 -> 7 -> 8 -> 13 -> 14 -> 6 -> 7 -> 8 -> 15 -> 16 -> 6 -> 7 -> 8 -> 17 -> 18 -> 6 -> 7 -> 8 -> 4 -> 5
BFS Path Found:
S -> 1 -> 5 -> G
```

Expanded Nodes List	Frontier List
S S1 S12 S123 S1234 S12345 S123456 S1234567 S12345678 S123456789 S12345678910 S123456791011	S 123 2345 345678 456789 567891011 67891011G12 7891011G121314 891011G1213141516 91011G12131415161718 1011G121314151617181 9 11G1213141516171819 G1213141516171819

3. (**20 points**) For the graph in Figure 1, implement **Depth First Search** (DFS) and complete the table below with the order in which the nodes will be expanded. Include the shortest path found by DFS. You must also include a screenshot of your code output in this document, in addition to filling in the table below. You are provided with a skeleton code in Python. You can also code it up your own way or in another language of your choice.

Expanded Nodes List	Frontier List
S S1 S14 S1410 S141011	S 321 3254 3251110 32511 3212G

```
DFS Expansion Order:
S -> 1 -> 4 -> 10 -> 11 -> 5
DFS Path Found:
S -> 1 -> 5 -> G
```

4. (**25 points**) For the graph in Figure 2, implement A* to find the shortest path between nodes S and G using the heuristic provided in Figure 2. Write the following: (i) is the heuristic admissible? (ii) the shortest path found by A* using this heuristic; (iii) solution cost of the found shortest path; (iv) the number of nodes expanded (including the start and goal nodes); and (v) time taken to solve. Include a screenshot of your code displaying the answers to (ii) - (v). You are provided with a skeleton code in Python. You will need to fill in the missing lines in the code. You can also choose to implement A* in a different programming language of your choice.

5. (**10 points**) For the graph in Figure 2, implement A* with the heuristic in Figure 3. Write the following: (i) is the heuristic admissible? (ii) the shortest path found by A* using this heuristic; (iii) solution cost of the found

Question 4:

i. The heuristic is admissible as the h value is never greater than the cost to reach the goal.

```
--- Simplified A* Search Results ---
(ii) Shortest path found: S -> 1 -> 4 -> G
(iii) Solution cost: 14
(iv) Number of nodes expanded: 7
(v) Time taken to solve: 0.0063 ms
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```

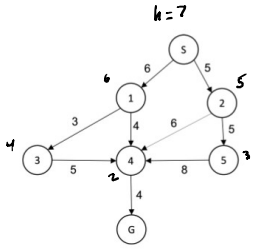

Node	h value
S	7
1	6
2	5
3	4
4	2
5	3
G	0

ii.

Exp.	Frontier
	S(7)
S	1(6+1), 2(5+5)
S2	1(12), 4(11+2), 5(10+3)
S21	~~4(13)~~, 5(13), 3(9+4), 4(10+2)
S214	5(13), 3(13), G(14)
S2145	3(13), G(14), 4(15+2)
S21453	G(14), 4(20), 4(14+2)
S214536	

Figure 2: Graph for informed search

shortest path; (iv) the number of nodes expanded (including the start and goal nodes); and (v) time taken to solve. Include a screenshot of your code displaying the answers to (ii) - (v).

Node	S	1	2	3	4	5	G
h value	10	9	1	1	4	1	0

Figure 3: Heuristic for Q5

6. (**15 points**) For the graph in Figure 2, (i) provide an admissible heuristic that is NOT consistent. (ii) Explain or prove how it satisfies admissibility and how it violates consistency. (iii) Solve the graph using your heuristic (either manually or by modifying the provided code to work with your proposed heuristic) and provide the solution path and cost. Does it find an optimal solution?

i./ii. if you change a heuristic like h(3) = 8 instead of 4, where the neighbor h(4)=2 and the cost to travel there is 5, 8 < 2+5 and is also not overestimating the actual cost of 5+4.

```
--- Simplified A* Search Results ---
(ii) Shortest path found: S -> 1 -> 4 -> G
(iii) Solution cost: 14
(iv) Number of nodes expanded: 7
(v) Time taken to solve: 0.0002 ms
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```

Still finds the optimal solution

Question 5:

i. it is inadmissible since h(1) = 9 > 8

```
--- Simplified A* Search Results ---
(ii) Shortest path found: S -> 1 -> 4 -> G
(iii) Solution cost: 14
(iv) Number of nodes expanded: 7
(v) Time taken to solve: 0.0002 ms
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```